Decision report: DEC-20261023-488023

EXP-CZLIFT-ed8113 · H-CZLIFT-cb95b8 · RQ-CZLIFT-b9f765 · GOAL-CZLIFT-1516d5 · 2026-10-23

Status

Written by the coordinator subagent, which has no shell. plot.py draws graph.svg from plot_data.json, and the session compiles and inspects report.pdf. The decision is not official until the ledger archive TASK-20261023-4d7449 passes post-commit verification and is pushed to the PR branch. That archive carries CORR-20261023-4ffd7e, EV-CZLIFT-ee9b23, DEC-20261023-488023, analysis.md, the H-CZLIFT-cb95b8 update and this folder.

Decision

- **Decision:** weaken. One transition: H-CZLIFT-cb95b8 approved -> weakened, attached to the statement as tested (CORR-20261023-4ffd7e).
- **Evidence strength:** preliminary (EV-CZLIFT-ee9b23). The red team's 'weak' maps to this level. The success reading depends on an undeclared sampler restriction. The rate is pigeonhole-dominated. The lift-specific part rests on one event measured outside the frozen protocol.
- **Claim tier:** toy (p <= 12809, 14 bits; r <= 3; relation bound 6).
- **Refutation basis:** proof_status certificate for the adverse clause only: an explicit tangent counterexample, re-found by the validator's independent blind search, verified exactly by the red team, and checked by hand in analysis.md I-1. The mechanism finding rests on a retrieved primary source (JKSST 2000).
- **Records:** EV-CZLIFT-ee9b23, DEC-20261023-488023, CORR-20261023-4ffd7e, H-CZLIFT-cb95b8, EXP-CZLIFT-ed8113, RUN-CZLIFT-c3f8a8 (seed 20261009), RUN-CZLIFT-218843 (seed 20261010), VAL-20261022-b08129 (TASK-20261022-67902e), RT-20261023-d796bb (TASK-20261023-62eb31), TASK-20261009-b8a2d5 (execution), TASK-20261023-4d7449 (ledger archive), DEC-20261009-296cee (approval), DEC-20261009-6aef7a (observation reading, not re-scored).
- **Knowledge promotion:** none. Not warranted: the decision is weaken on preliminary evidence, and its durable content is a toy re-measurement already covered by JKSST 2000 section 5.

Exact scoped claim

Setting: p in {101, 211, 401, 809, 1601, 3203, 6421, 12809}; random y^2 = x^3 + a x + b over F_p; r in {2, 3} points drawn uniformly from the affine points with y != 0 and pairwise distinct x (the hash-pinned sampler harness/czlift.py random_point); symmetric integer lifts; the unique zero-freedom rational Weierstrass fit; implementation experiments/EXP-CZLIFT-ed8113/implementation/run.py at run commit 025949c406; about 200 s per run against a 3600 s advisory budget. Result: the exact search over max |m_i| <= 6 found 1 relation in 3200 valid trials (EV-CZLIFT-ee9b23; RUN-CZLIFT-c3f8a8 and RUN-CZLIFT-218843). The rate is 3.1e-4, with exact 95 percent CI [7.9e-6, 1.7e-3]. Per seed: 1/1600 (one-sided upper 0.00296) and 0/1600 (0.00187). For r = 3: 0/1600. The one relation is the tangent relation 2 P~1 + P~2 = O at p = 101, r = 2. The planted (2, -1) control was detected 48/48. No dependence rate above the predefined 0.01 falsification threshold was observed over the tested primes, sampler, fit, relation bound, implementation and budget. The predefined zero-count prediction was not met on seed 20261009.

Not claimed: 'never dependent'; 'independent' (read from a bounded search); the xedni obstruction or its counting form; 'random points of E(F_p)' without the y != 0 restriction; any statement about xedni proper (r = 4..9, discriminant minimization, reverse-Mestre conditions), index calculus, cost or any deployed curve.

Observation

Validity, checked first. 2 of 2 planned trials ran; both are output_validated with returncode 0 and check_returncode 0. The run commit 025949c406 is clean and hash-identical to the freeze 172ec3c94f. 9/9 artifact hashes per run match. Every frozen metric recomputes exactly (validator j2_recompute.py). A sealed blind replay agrees on 3200/3200 trial outcomes and 48/48 controls. There are 0 singular exclusions in 32 cells. Both runs declare certificate kind none (measurement runs). The one recorded dependence was re-verified by independent code. The run set is admissible.

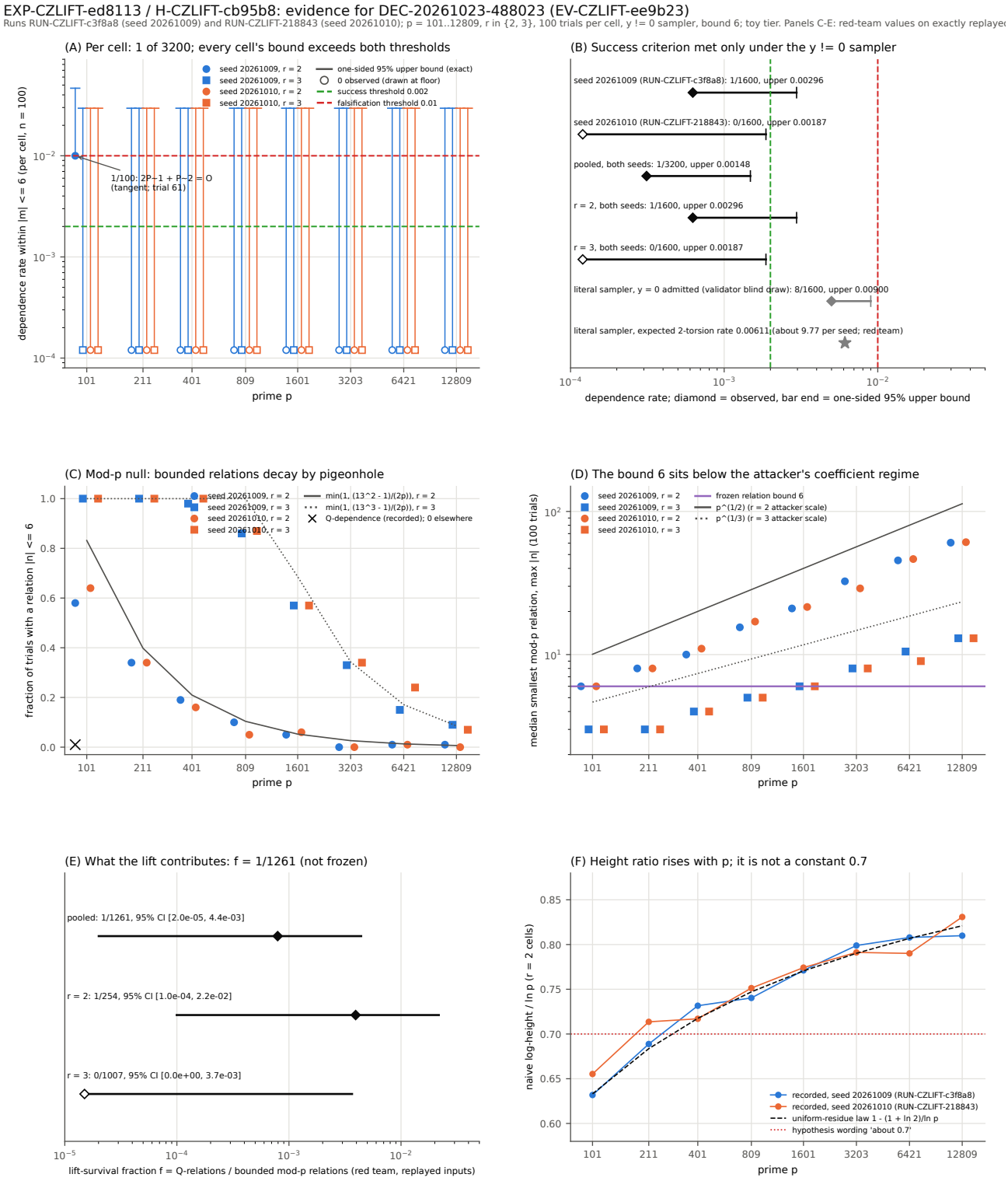


Figure 1: Panel A: dependence rate per (p, r, seed) cell, n = 100 per cell, with exact one-sided 95 percent upper bounds (open marker = 0 observed), against the frozen 0.002 success and 0.01 falsification thresholds (recorded; RUN-CZLIFT-c3f8a8, RUN-CZLIFT-218843). Panel B: per-seed, pooled and per-r rates, the validator's literal-sampler blind draw (8/1600), and the red team's literal-sampler expectation (0.0061). Panel C: mod-p bounded-relation rate on the same residues against (13^r - 1)/(2p), with the Q-dependence rate. Panel D: median smallest mod-p relation coefficient per cell against the bound 6 and p^(1/r). Panel E: lift-survival fraction with exact 95 percent intervals. Panel F: recorded height ratio (r = 2 cells) against 1 - (1 + ln 2)/ln p and the hypothesis' 'about 0.7'. Panels C-E are red-team values on exactly replayed inputs (rt_modp_null.json, rt_stats.json), not frozen metrics. Data: plot_data.json; drawn by plot.py.

Recorded: seed 20261009, 1 dependent in 1600 (0.000625; r = 2 0.00125, r = 3 0.0); seed 20261010, 0 in 1600; controls 24/24 on each seed. The dependence is at p = 101, r = 2, trial 61: lifts (-24, -15) and (48, -15) on Y^2 = X^3 - 1728 X - 27423, relation [-2, -1]. Height ratio 0.632-0.810 and 0.655-0.831, rising with p. Reviewer values, read by key. The sampler never draws y = 0 (0 of 60000; 0 of 8000 replayed points). The height-pairing determinant shows 3199/3200 tuples Z-independent. There are 1261/3200 bounded mod-p relations, of which 1 lifts (f = 7.9e-4, CI [2.0e-5, 4.4e-3]). The median smallest mod-p relation runs from 6 to 60.5-61 (r = 2) and from 3 to 13 (r = 3).

The JKSST empirical law expects 0.855 dependent trials. In-distribution order-7 and order-8 torsion pairs at p = 6421 pass the bound-6 search.

Comparison

- Frozen success criterion (<= 0.002 on both seeds, controls 1.0): met. Frozen falsification criterion (> 0.01 on both seeds): not met. No invalidation rule fired. Nothing is re-scored, and no amendment is made.
- Preregistered point prediction (0 of 1600 per seed, upper bound 0.0019): met on seed 20261010 and not met on seed 20261009. The statement's 'upper 95 percent bound below 0.002' holds pooled (0.00148), not on seed 20261009 (0.00296).
- Sampler: under the literal all-points reading, 2-torsion hits alone give about 9.77 dependent trials per seed (0.0061), between the two thresholds; P(success on both seeds) = 1.5e-4. The validator's literal draw gave 8/1600.
- Against expectation: the validator's exact tangent-family rate gives P(at least one hit in 200 r = 2 trials at p = 101) = 0.179; the JKSST law gives 0.855 expected; 1 was observed.
- Against the mod-p null: the Q-count is bounded by the mod-p count, which falls from 58-64 percent (r = 2, p = 101) to 0-1 percent (p >= 6421). The large-p zero is forced by pigeonhole.
- Against the attacker: bound 6 lies below the median smallest mod-p relation in every r = 2 cell with p >= 211 and every r = 3 cell with p >= 3203. The attacker's scale is p^(1/r).

Inference

Counterexample (analysis.md I-1). On Y^2 = X^3 - 1728 X - 27423, both (-24, -15) and (48, -15) lie on the curve, since each gives 225 = 15^2. The tangent at (-24, -15) is Y = -15, because X^3 - 1728 X - 27648 = (X + 24)^2 (X - 48). Hence 2 P~1 + P~2 = O. The doubling slope of 2 P~1 = (48, 15) is 864/5, so 4 P~1 is non-integral, and P~1 has infinite order (Nagell-Lutz). The zero clause and the zero prediction are false on seed 20261009; nothing else is refuted by this instance. Mechanism. JKSST 2000 (retrieved) bounds xedni success by C0/p conditionally on Lang's height conjecture and log|D| >= C1 max hhat. The bound is vacuous at p <= 12809 (C0 ~ 1e4 for r = 2), and the same paper measures frequent dependence at small discriminants. It does not give dependence as negligible at this scale. The bound-6 search measures the regime an attacker cannot use. It cannot certify independence. Reading. A preregistered prediction failed, by certificate, and the mechanism is wrong in two sentences. The broad rarity reading survives at toy scale, consistent with the literature, and the frozen falsification criterion did not fire. That is weaken, not reject_scoped (a miscalibrated point prediction is not a refuted mechanism) and not support (restating to fit the outcome and then supporting would be circular).

Obstruction (EV-CZLIFT-ee9b23). (q1) Q-dependence within bound 6: 1/3200 = 3.1e-4 [7.9e-6, 1.74e-3]. (q2) Mod-p bounded-relation rate 1261/3200, decaying like (13^r - 1)/(2p). (q3) Lift-survival f = 7.9e-4 [2.0e-5, 4.4e-3]. (q4) Median smallest mod-p relation from 6 to 61 (r = 2) and 3 to 13 (r = 3), against bound 6. Measured by RUN-CZLIFT-c3f8a8 and RUN-CZLIFT-218843 (q1); q2-q4 by the red team on exactly replayed inputs. Scope: the eight primes, r in {2, 3}, the y != 0 sampler, symmetric lifts, the zero-freedom fit, bound 6, toy. Resource check (examined): per-round success is Pr[bounded mod-p relation] x f with Pr <= (2 C2)^r / n, so no f in [0, 1] gives a sublinear attack. Rho dominates, and lifted relations already hold mod p. C1 >= 11.24 is the hypothesis of JKSST Lemma 4.1 and holds. No theory takes these as a resource; nothing is spawned.

Limitation

- Toy tier only: p <= 12809, r <= 3, bound 6, symmetric lifts, zero-freedom fits, y != 0 sampler. Nothing about xedni proper, index calculus on E(Q), cost or any deployed curve.
- The literal sampler was not tested by the producer runs. The lift-survival fraction, the unbounded independence count and the coefficient medians are reviewer measurements on replayed inputs, and PARI is unpinned.
- One relation in 3200 trials. Per-cell resolution is coarse: the one-sided upper bound is 0.0295 at 0/100, above both thresholds.
- The singular-fit control is vacuous, and the planted control covers only max |m| = 2, r = 2 (the validator's edge probe covers |m| = 6, r = 3). check.py imports run.py. The height metric covers r = 2 cells only. There is no memory record.
- The JKSST empirical law is applied outside its fitted regime. The composing Coordinator has no shell and checked only the I-1 identities itself.

Review inputs

- reports/validation-20261022/validation_report.yaml (VAL-20261022-b08129: J1, J2, J3, PTM-1 hold; findings F1-F5)
- reports/redteam-20261023/redteam_report.yaml (RT-20261023-d796bb: R1, R2, PTM-1 break; narrowest supported statement; rt_* artifacts)
- ledger/corrections/CORR-20261023-4ffd7e.yaml (statement as tested; criterion reading)
- No earlier reports/run-* directory exists for this experiment; the run records are the manifests under runs/.

Canonical graphs checked

knowledge/frontiers/ecdlp/index-calculus/KR-IC-cd159a.yaml (xedni and lifting to characteristic zero) was checked. It is a frontier row, not a graph, and this decision leaves it unchanged. Its 'negligible probability, status proven' wording rests on abstracts and overstates the conditional primary result; that is routed to the synthesis task (DEC-20261023-488023 N-4b). No canonical graph under knowledge/frontiers/ecdlp/ plots this quantity.

Next actions

1. Session (N-1): write analysis.md from the coordinator's returned text, run plot.py (its bound check must pass), and compile and inspect report.pdf. Run tools/validate_ledger.py and tools/check_review_independence.py. Then run the ledger archive TASK-20261023-4d7449 alone by exact path. Verify, merge origin/main, push, and open or refresh the ready PR 'evidence: EV-CZLIFT-ee9b23/DEC-20261023-488023'.

2. N-2: no follow-up, successor or replication on H-CZLIFT-cb95b8. The red team's 1-CPU-minute replay (height-pairing determinant and lift-survival as preregistered metrics) is weighed and not dispatched. On these inputs both values are already observed, so a prediction would be in-sample. No value of f changes any decision. Its method becomes contract requirements (N-5) on fresh seeds.
3. N-3 (synthesis task on RQ-CZLIFT-b9f765): the bounded count is not 'the xedni obstruction stated as a count'; EXP-CZLIFT-ed8113 cannot close the coverage clause of decision_target (a); name the point distribution wherever 'random points' appears; require the mod-p null, lift-survival and an unbounded certificate for any bounded relation search; mark Silverman-Suzuki and Huang-Kueh-Tan as recalled; reserve 'xedni' for xedni proper.
4. N-4 (same task): a correction on the EXP-CZLIFT-ed8113 specification (sampler, r = 2 height metric, prediction-versus-criterion inconsistency, on_positive not invoked); a superseding row or correction for KR-IC-cd159a.
5. N-5: binding requirements for future lifting contracts. Stated sampler; unbounded independence certificate with PARI pinned; mod-p null and lift-survival frozen; a calibrated null instead of point-zero; any coefficient bound >= 12; Lang and Lemma 4.1 numbered as heuristics where load-bearing; an independent checker; per-trial inputs recorded.
6. N-6: a xedni-proper nearby-object arm (discriminant-minimized fits, r = 4..6) is a ranking candidate only, with expected rank low. N-7: GOAL-CZLIFT-1516d5 stays active with exactly one next action. Keep 'compose the remaining evidence decisions of this round' if it is open; otherwise dispatch the RQ-CZLIFT-b9f765 synthesis task.